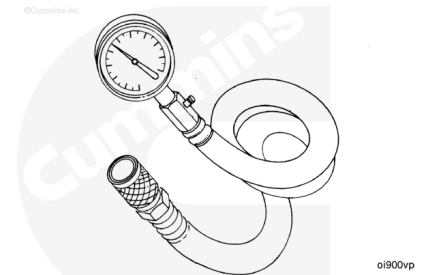


Trainings ▾

Initial Check

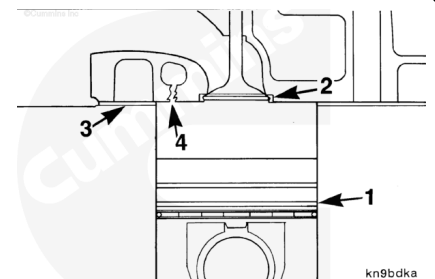
Compression Gauge and Adapter

It is very time consuming and expensive to begin removing internal engine components to diagnose failures. A compression gauge can be used as an aid to check for proper functioning of the following components:



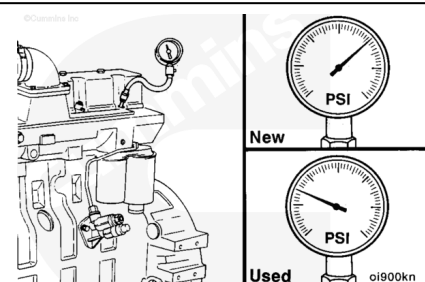
1. Piston ring sealing
2. Intake and exhaust valve sealing
3. Cylinder head gasket sealing
4. Cylinder head cracked.

See the appropriate procedure in this manual for replacement of failed components.



Note : Due to variables such as starter and battery conditions that affect engine cranking speed, it is difficult to establish an absolute value for compression pressure; however, the following values can be used as guidelines:

- New engine (cranking speed at 250 rpm) 2413 kPa [350 psi]
- Used engine (cranking speed at 250 rpm) 2068 kPa [300 psi].

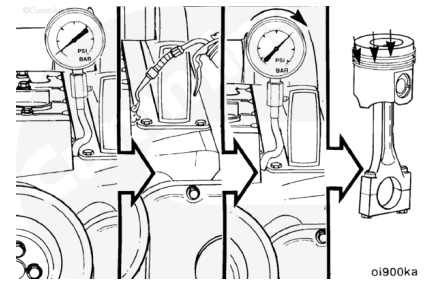


It is recommended that the compression pressure be checked on all cylinders and then compared. All cylinders should be within 690 kPa [100 psi] of each other.

Piston Ring Sealing

If the compression is low but can be increased significantly by squirting oil into the cylinder, the cause of low compression is inadequate sealing between the rings and the cylinder walls. Use the following procedure for piston ring replacement.

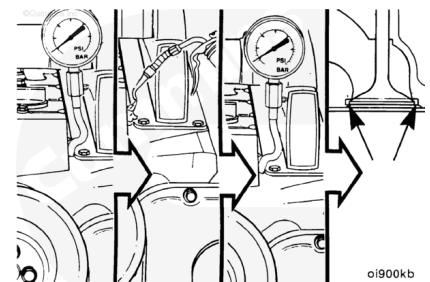
- Use the following procedure for C Series engines. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-047.html>)
- Use the following procedure for C8.3G, C Gas Plus, and L Gas Plus engines. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/39/39-001-047.html>)



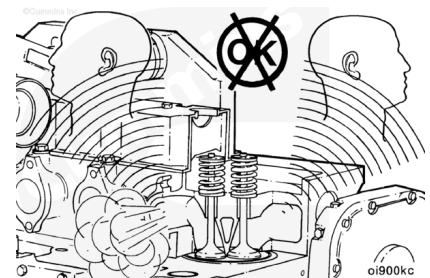
Intake and Exhaust Valve Sealing

If the compression is low on one or more nonadjacent cylinders and the pressure can **not** be increased by oiling the rings, poor valve sealing is to be suspected.

- Use the following procedure for C Series engines. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html>)
- Use the following procedure for C8.3G, C Gas Plus, and L Gas Plus engines. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/39/39-002-004-tr.html>)



Valve leakage is often audible from the intake and exhaust manifold.

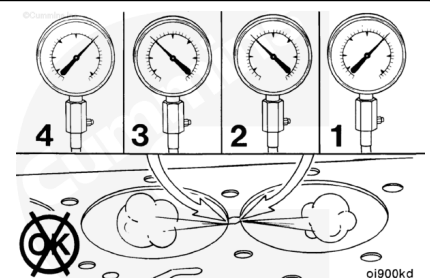


Cylinder Head Gasket Sealing

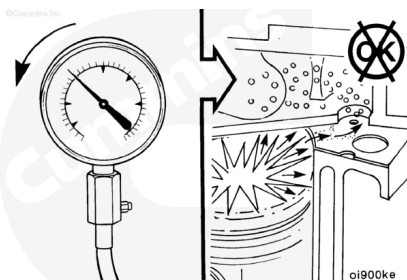
If the compression is low on adjacent cylinders and the pressure can **not** be increased by oiling the rings, the cylinder head gasket is probably leaking between the cylinders.

Refer to Procedure 002-021 in Section 2.

(</qs3/pubsys2/xml/en/procedures/41/41-002-021.html>)

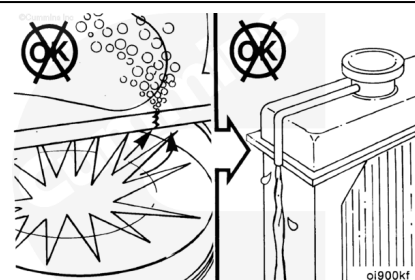


Low compression on a single cylinder can be caused by an external leak or a leak to a coolant passage. A leak to a coolant passage of this magnitude will also result in coolant in the cylinder.



A compression leak to the coolant will normally be detected by loss of coolant as the coolant is blown from the cooling system.

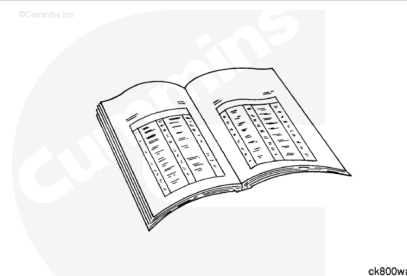
Service Tip: Remove the drive belt from the water pump. Run the engine for 1 to 2 minutes and check for coolant being blown from the radiator by compression gases.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Close the fuel supply valve. Refer to the OEM service manual.
- Disconnect the batteries. Refer to the OEM service manual.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/100/100-007-037.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/100/100-007-025.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html>)
- Remove the cylinder head.
 - Use the following procedure for C Series engines. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html>)
 - Use the following procedure for C8.3G, C Gas Plus, and L Gas Plus engines. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/39/39-002-004-tr.html>)
- Remove and disassemble the piston and connecting rod assemblies.

- Use the following procedure for C Series engines. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/41/41-001-054.html)
- Use the following procedure for C8.3G, C Gas Plus, and L Gas Plus engines. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/39/39-001-054-tr.html)
- Remove the piston rings.
 - Use the following procedure for C Series engines. Refer to Procedure 001-047 in Section 1.
(/qs3/pubsys2/xml/en/procedures/100/100-001-047.html)
 - Use the following procedure for C8.3G, C Gas Plus, and L Gas Plus engines. Refer to Procedure 001-047 in Section 1.
(/qs3/pubsys2/xml/en/procedures/39/39-001-047.html)

Clean and Inspect for Reuse

⚠ WARNING ⚠

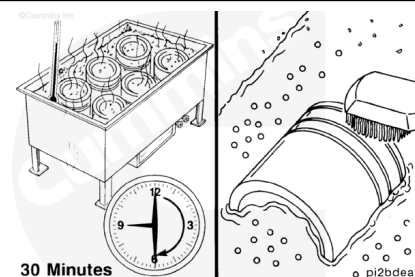
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

Make sure the cleaning solvent is approved for aluminum. Damage to the pistons can result.

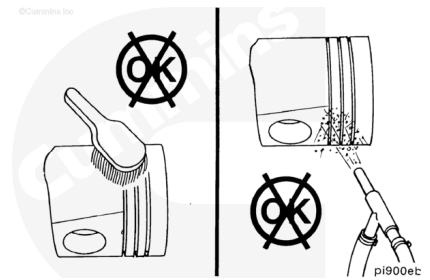


Allow the pistons to soak for a minimum of 30 minutes in a tank containing an approved cleaning solvent for aluminum.

Use a hot, soapy solution and nonmetallic brush to remove carbon deposits.

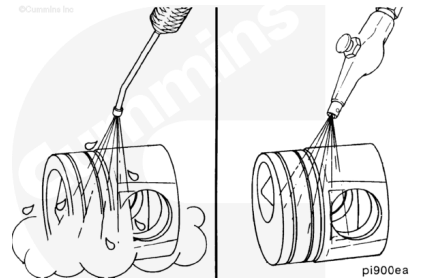
⚠ CAUTION ⚠

Do not use a metal brush. A metal brush will damage the piston ring grooves. Do not use glass beads, walnut shells, or plastic bead blasting to clean the piston ring grooves or pin bores. This can cause the pin bore to crack or the rings not to seat.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

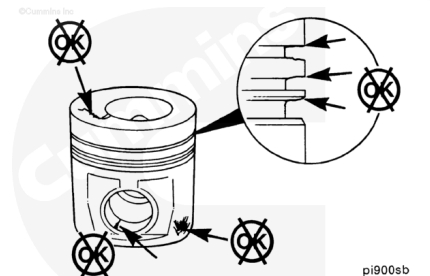
Use steam to clean the pistons.

Dry with compressed air.

Note : Dimensional inspection of the piston is required only when the cause of the lack of sealing is not apparent.

Inspect the piston for damage and excessive wear. Check the top ring grooves, skirt, and pin bore.

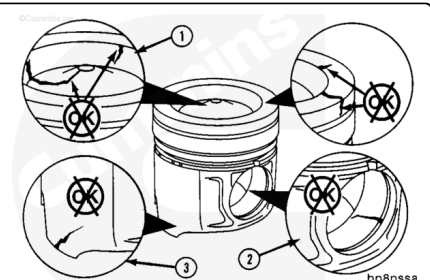
Inspect the piston for rolled edges on the ring lands. If the ring lands are rolled in, this indicates ring beat-in and can cause loss of sealing.



Inspect the piston bowl (1), pin bore (2), and skirt (3) for cracks or damage.

Note : Do not use pistons with cracks.

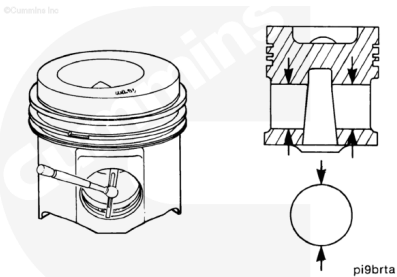
Note : If severe piston damage has occurred, check the turbocharger and other exhaust components for



damage from debris.

Measure the piston bore when the piston temperature is at 20°C [68°F].

Piston Pin Bore Inside Diameter		
mm		in
45.01	MIN	1.772
45.03	MAX	1.773



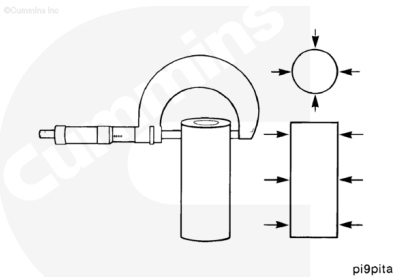
Note : Add 0.013 mm [0.0005 in] to the piston pin bore inside diameter per 5°C [10°F] temperature rise up to 32°C [90°F].

The piston pin bore **must** be checked closer to the outside diameter of the piston since the piston pin bore is tapered.

Inspect the piston pin for scratches, grooves, or other damage.

Measure the piston pin outside diameter.

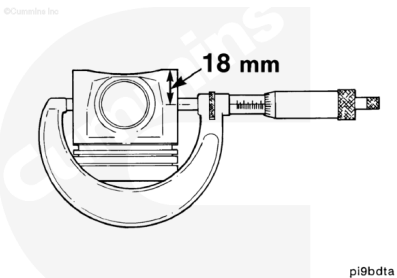
Piston Pin Diameter		
mm		in
44.99	MIN	1.771
45.00	MAX	1.772



Note : Discard the piston pin if it is more than 0.03 mm [0.001 in] out of round.

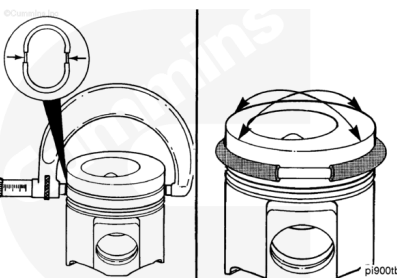
Measure the piston skirt diameter as illustrated.

Piston Skirt Diameter		
mm		in
113.81	MIN	4.481
113.88	MAX	4.484



Use a micrometer and piston ring groove gauge, Part Number 3823966, to measure the wear of the top ring groove keystone angle. Use piston ring groove gauge, Part Number 3823965, to measure the wear of the intermediate ring groove keystone angle.

Piston Ring Groove (Keystone Angle)

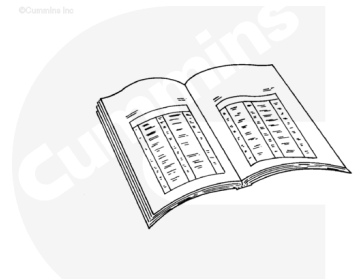


	mm		in
Top Ring:	113.94	MIN	4.486
Intermediate Ring:	114.32	MIN	4.501

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

- Install the piston rings.
 - Use the following procedure for C Series engines. Refer to Procedure 001-047 in Section 1.
(/qs3/pubsys2/xml/en/procedures/100/100-001-047.html)
 - Use the following procedure for C8.3G, C Gas Plus, and L Gas Plus engines. Refer to Procedure 001-047 in Section 1.
(/qs3/pubsys2/xml/en/procedures/39/39-001-047.html)
- Assemble and install the piston and connecting rod assemblies.
 - Use the following procedure for C Series engines. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/41/41-001-054.html)
 - Use the following procedure for C8.3G, C Gas Plus, and L Gas Plus engines. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/39/39-001-054-tr.html)
- Install the cylinder head.
 - Use the following procedure for C Series engines. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html)

- Use the following procedure for C8.3G, C Gas Plus, and L Gas Plus engines. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/39/39-002-004-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-025.html)
- Fill the engine with clean lubricating oil approved for natural gas engines. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-037.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Open the fuel supply valve. Refer to the OEM service manual.
- Operate the engine and check for leaks.